

Random Walks

- **Final project details**
- **Stock Option pricing**
- **Black–Scholes(–Merton) Equation**
- **AM cake today (every Thursday) in
Pierce 307B, 2:40pm**

Final project

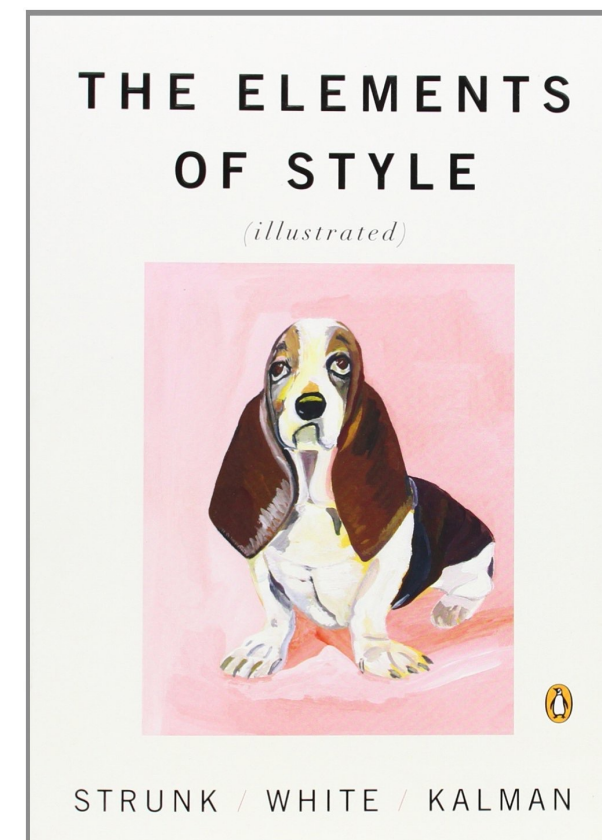
- **Worth 30% of the course grade**
- **May be done with a partner or individually**
- **Graded out of 80 points**
- **Deadline: 5pm on May 3 (last day of Reading Period)**

What are we looking for?

- We are looking for you to **Apply Mathematics**
- **“Apply”** – we want you consider some real-world question and not just something purely abstract. You should think about how you can mathematically model the real world.
- **“Mathematics”** – we want your project to have some scope for doing some significant mathematics.
- Another viewpoint: what question are you trying to answer and why would other people care about it?
- While many projects involve some programming, purely analytical projects are also allowed.

Length guideline

- For text and figures:
 - 7 pages for individual project
 - 12 pages for pair project
- This is a *highly approximate* guideline. The exact length is not important. Depending on the topic, some projects may be longer, and some may be shorter. The key issue is how much did you accomplish.
- Text should be clear and concise: see *The Elements of Style* by Strunk, White, and Kalman



Project proposal

- **Consider project ideas over the next few weeks. Your project could relate to one of the topics covered in class, or could be about a completely different subject. Please discuss with the teaching staff.**
- **Project proposal due by April 6th at 5 PM, and consists of a half-hour meeting with a Margo.**
- **May optionally submit a half-page overview of project prior to meeting**
- **5 points automatically awarded for doing this.**
- **Meetings arranged by appointment. Good days:**
3/8, 3/9, 3/10, 3/27, 3/28, 3/29, 3/30, 3/31, 4/3

Grading breakdown: content

Topic	Points
Project proposal completed by April 6th, 5 PM	5
Project motivation: what problem are you trying to solve? What has been done before in this area? If appropriate, cite any relevant books and papers.	10
Project methods and results: what mathematics and/or code did you develop for your problem? Where appropriate, did you consider mathematical analyses of your approach? Is the mathematics/code that you developed correct?	26
Project conclusions: did you solve what you set out to do? What are possible limitations and problems with your approach? How could you develop the project further?	9

Total for content: 50 points

Grading breakdown: style

Topic	Points
Written material: clear, correct grammar and spelling, appropriate citations	30
Figures (if any): clear, properly labeled	
Code listings (if any): clear, well commented	

Total for style: 30 points

Grand total (content + style): 80 points

Rough guideline for layout

- **Abstract:** a short, concise summary of your paper
- **Introduction:** background information about your project and your original motivation. References to any relevant papers.
- **Methods:** description of all of the methods (mathematics, coding, *etc.*) that you used
- **Results and discussion:** Explain your results, including your graphs, why they are important and why we should care. Critically examine the quality of your results and the quality of your models.
- **Conclusion:** a quick round-up of the main points that you learned, plus possible future work
- **References**

Option Pricing

The Pricing of Options and Corporate Liabilities

1973



Fischer Black

University of Chicago

Myron Scholes

Massachusetts Institute of Technology

If options are correctly priced in the market, it should not be possible to make sure profits by creating portfolios of long and short positions in options and their underlying stocks. Using this principle, a theoretical valuation formula for options is derived. Since almost all corporate liabilities can be viewed as combinations of options, the formula and the analysis that led to it are also applicable to corporate liabilities such as common stock, corporate bonds, and warrants. In particular, the formula can be used to derive the discount that should be applied to a corporate bond because of the possibility of default.

Jargon

Stock – share of a company

Derivative – broad financial term to describe financial instrument whose value is derived from that of other assets, *e.g.*

- **stocks**
- **bonds**
- **commodities (oil prices)**
- **other indices (weather, *etc.*)**

Primarily used for managing risk, or hedging

Jargon

Call option – The right, but not the obligation, to **buy some asset** in the future at a fixed price that is agreed upon today, called the **strike** or exercise price.

Put option – The right, but not the obligation to **sell an asset** in the future at a price determined today.

The person **buying the option** is said to be in a *long* position: if they keep the option they won't make money for awhile.

The person **selling the option** is said to be in a *short* position.

An option is a security giving the right to buy or sell an asset, subject to certain conditions, within a specified period of time. An “American option” is one that can be exercised at any time up to the date the option expires. A “European option” is one that can be exercised only on a specified future date. The price that is paid for the asset when the option is exercised is called the “exercise price” or “striking price.” The last day on which the option may be exercised is called the “expiration date” or “maturity date.”

The simplest kind of option is one that gives the right to buy a single share of common stock. Throughout most of the paper, we will be discussing this kind of option, which is often referred to as a “call option.”

Jargon

5 Minute Break

Explain the jargon to your neighbor

Example – Call Option

- You have a June call option for Microsoft with a strike price of \$30 for 100 shares. If the price today is \$20 and it rises to \$40, would you exercise your option?
 - **Yes**, you would exercise your call option and buy Microsoft stock for \$30, earning \$10 in profit
 - The seller writes the call option contract and the owner (buyer of contract) has the option to buy the said stock at the given value
 - Note: seller is required to sell, but buyer isn't required to buy
 - The seller must give the buyer \$10
- Hence, the option contract costs money; the price depends on the strike price, the length of time, risk, *etc.*

Option Pricing

- **Note the option is an insurance contract. The question of valuing options is really thus the question of how much to charge for insurance.**
- **How do we value this?**
- **We do not know what is going to happen to the stock price**

Example – Call Option

You have a June call option for Microsoft with a strike price of \$30 for 100 shares. Suppose the price today is \$20 and, as of today, there is a 1/3 chance each that the stock will go up to \$27, to \$33, or to \$39.

How much should you charge for the option?

5 Minute Break

(Solve this problem with your neighbor)

How much money do you make?

- **If stock is \$27, the option is worthless. PAYOFF = 0**
- **If stock is \$33, the option is worth \$3. PAYOFF = \$3**
- **If stock is \$39, the option is worth \$9. PAYOFF = \$9**
- **Expected payoff = $\frac{1}{3} * 0 + \frac{1}{3} * 3 + \frac{1}{3} * 9 = \4**

Thus you should charge *at least* this much for the option

Example – Call Option

You have a June call option for Microsoft with a strike price of \$30 for 100 shares. Suppose the price today is \$20, and as of today, there is a 1/3 chance **each that the stock will go up to \$30, to \$40, or to \$50.**

Does the value of the option change?

How much money do you make?

- If stock is \$30, the option is worthless. PAYOFF = 0**
- If stock is \$40, the option is worth \$10. PAYOFF = \$10**
- If stock is \$50, the option is worth \$20. PAYOFF = \$20**
- Expected payoff = $1/3 * 0 + 1/3 * 10 + 1/3 * 20 = \10**

The value of the option should change. (The expected value of the stock also changes—it is now \$40/share.)

Example – Call Option

You have a June call option for Microsoft with a strike price of \$30 for 100 shares. If the price today is \$20, and as of today, there is a 1/3 chance **each that the stock will go up to \$21, to \$40, or to \$59.**

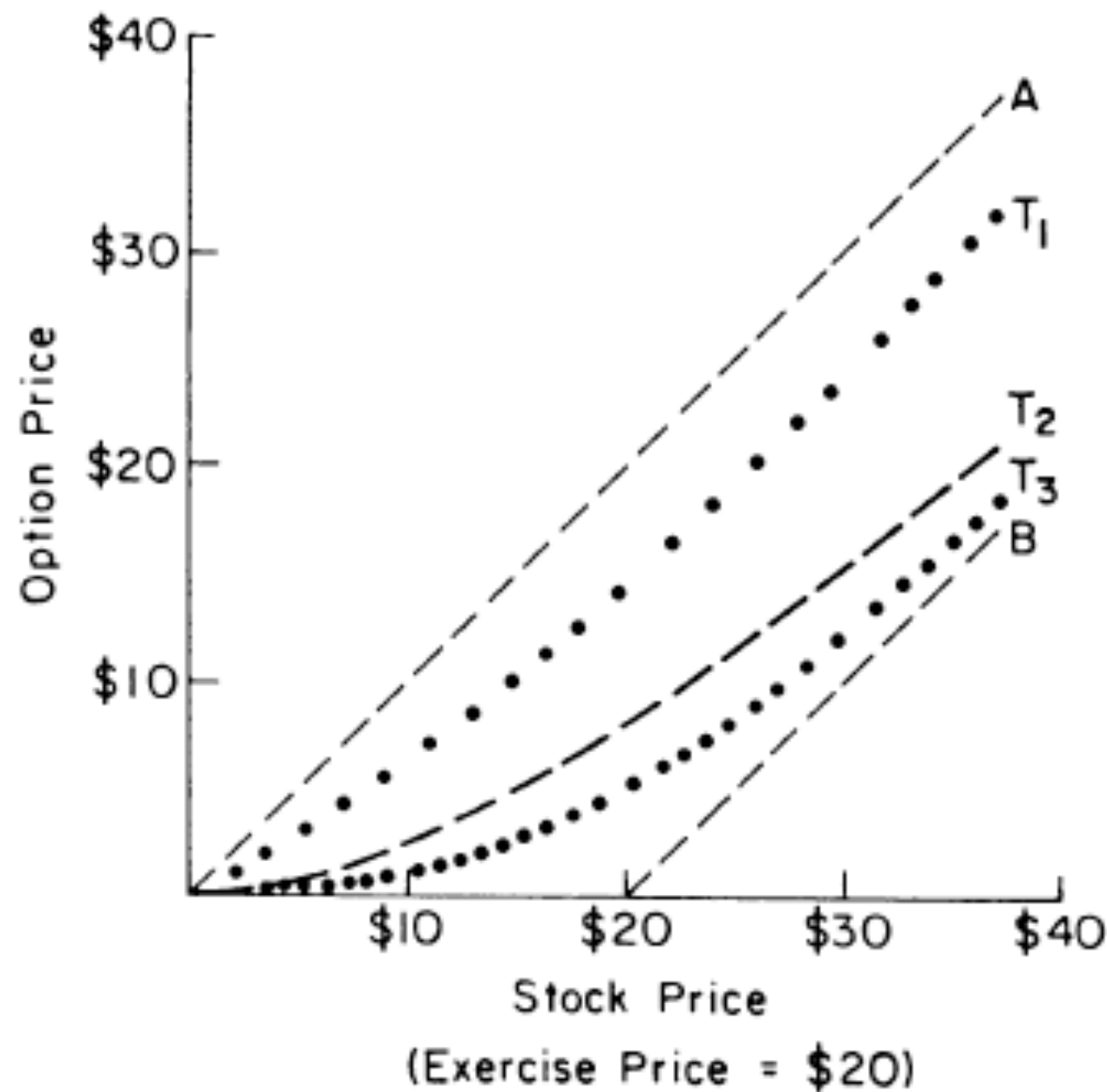
Does the value of the option change?

How much money do you make?

- If stock is \$21, the option is worthless. PAYOFF = 0**
- If stock is \$40, the option is worth \$10. PAYOFF = \$10**
- If stock is \$59, the option is worth \$29. PAYOFF = \$29**
- Expected payoff = $1/3 * 0 + 1/3 * 10 + 1/3 * 29 = \13**

The value seems to depend both on the expected value of the stock and on the volatility

Option Price vs. Stock Price



Line A

Maximum value of option

Line B

Minimum value of option

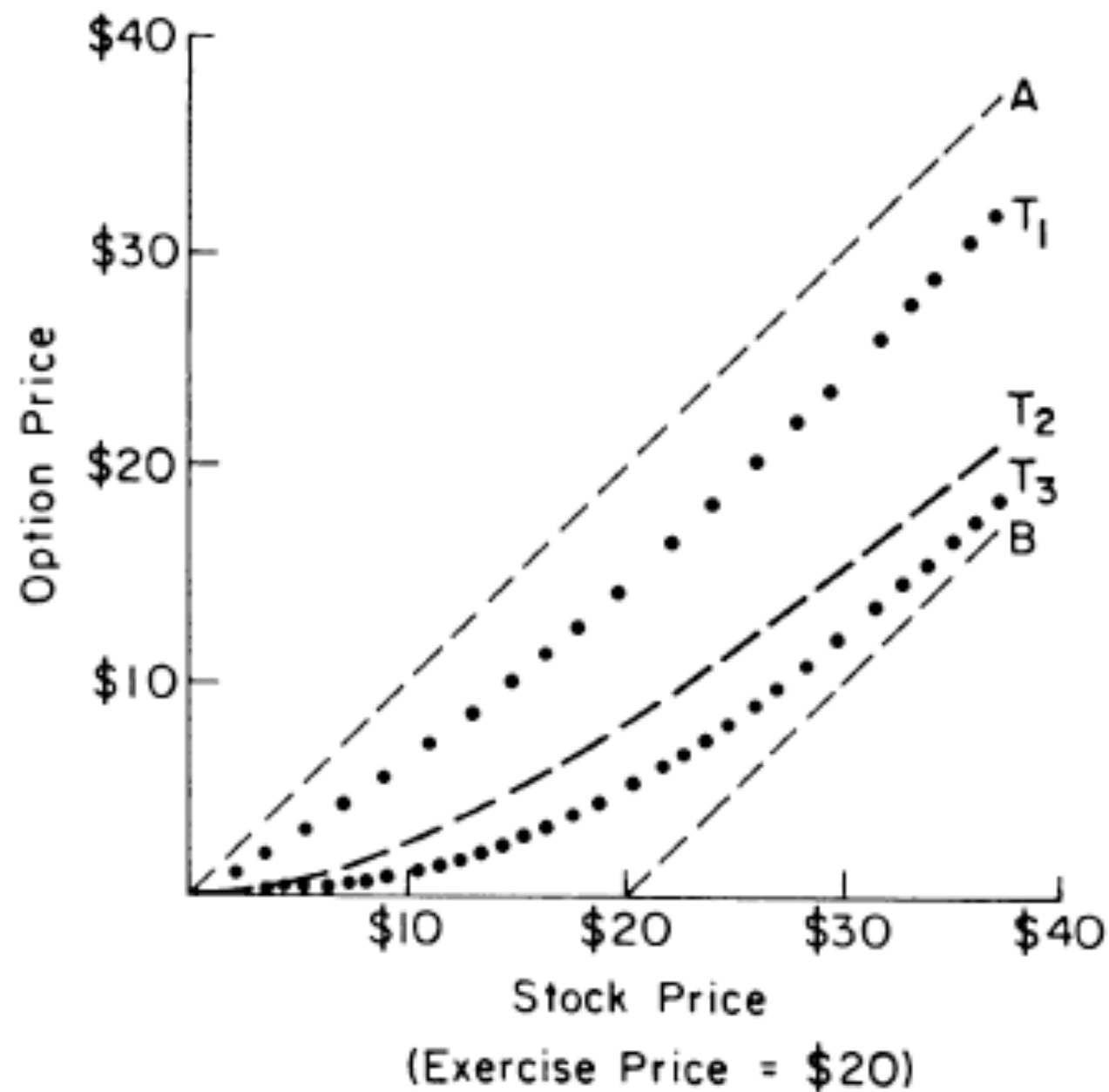
T₁, T₂, T₃

values at successively shorter maturities.

FIG. 1.—The relation between option value and stock price

Note: Exercise Price = Strike Price

Option Price vs. Stock Price



Why does the option value approach line B as the time to maturity shrinks?

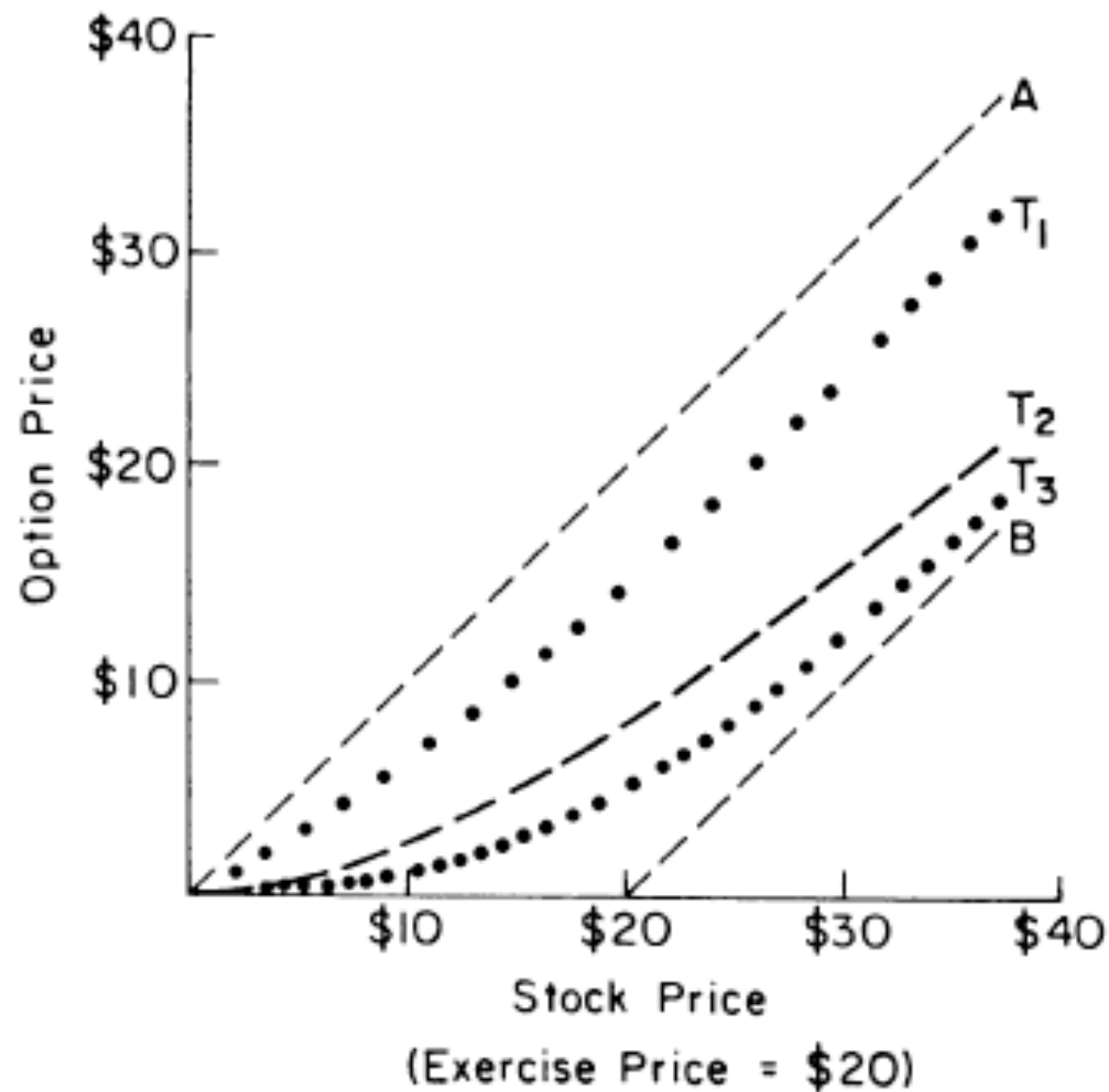
Intuitively, what determines where it starts (A) and ends (B)?

5 minute break to discuss with your neighbor

FIG. 1.—The relation between option value and stock price

Note: Exercise Price = Strike Price

Which is more volatile?



Suppose I have:

- **\$100 in stock**
- **\$100 in options**

Which is more likely to change in value?

FIG. 1.—The relation between option value and stock price

Note: Exercise Price = Strike Price

Other formulas, before Black Scholes

For example, Sprenkle's formula for the value of an option can be written as follows:

$$kxN(b_1) - k^*cN(b_2)$$
$$b_1 = \frac{\ln kx/c + \frac{1}{2}v^2(t^* - t)}{v\sqrt{(t^* - t)}}$$
$$b_2 = \frac{\ln kx/c - \frac{1}{2}v^2(t^* - t)}{v\sqrt{(t^* - t)}}$$

In this expression, x is the stock price, c is the exercise price, t^* is the maturity date, t is the current date, v^2 is the variance rate of the return on the stock,¹ $\ln$ is the natural logarithm, and $N(b)$ is the cumulative normal density function. But k and k^* are unknown parameters. Sprenkle (1961) defines k as the ratio of the expected value of the stock price at the time the warrant matures to the current stock price, and k^* as a discount factor that depends on the risk of the stock. He tries to estimate the values of k and k^* empirically but finds that he is unable to do so.



Paul Samuelson



Robert C. Merton

More typically, Samuelson (1965) has unknown parameters α and β , where α is the rate of expected return on the stock, and β is the rate of expected return on the warrant or the discount rate to be applied to the warrant.² He assumes that the distribution of possible values of the stock when the warrant matures is log-normal and takes the expected value of this distribution, cutting it off at the exercise price. He then discounts this expected value to the present at the rate β . Unfortunately, there seems to be no model of the pricing of securities under conditions of capital market

In a subsequent paper, Samuelson and Merton (1969) recognize the fact that discounting the expected value of the distribution of possible values of the warrant when it is exercised is not an appropriate procedure. They advance the theory by treating the option price as a function of the stock price. They also recognize that the discount rates are determined in part by the requirement that investors be willing to hold all of the outstanding amounts of both the stock and the option. But they do not make use of the fact that investors must hold other assets as well, so that the risk of an option or stock that affects its discount rate is only that part of the risk that cannot be diversified away. Their final formula depends on the shape of the utility function that they assume for the typical investor.

The Valuation Formula

In deriving our formula for the value of an option in terms of the price of the stock, we will assume “ideal conditions” in the market for the stock and for the option:

- a)* The short-term interest rate is known and is constant through time.
- b)* The stock price follows a random walk in continuous time with a variance rate proportional to the square of the stock price. Thus the distribution of possible stock prices at the end of any finite interval is log-normal. The variance rate of the return on the stock is constant.
- c)* The stock pays no dividends or other distributions.
- d)* The option is “European,” that is, it can only be exercised at maturity.
- e)* There are no transaction costs in buying or selling the stock or the option.
- f)* It is possible to borrow any fraction of the price of a security to buy it or to hold it, at the short-term interest rate.
- g)* There are no penalties to short selling. A seller who does not own a security will simply accept the price of the security from a buyer, and will agree to settle with the buyer on some future date by paying him an amount equal to the price of the security on that date.

**Discuss these assumptions with your neighbor.
Decide which one bothers you the most.**

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This assumption will show up in lab and homework

Hedging

$$w_1 = \frac{\partial w}{\partial x}$$

the values of known constants. Writing $w(x, t)$ for the value of the option as a function of the stock price x and time t , the number of options that must be sold short against one share of stock long is:

$$1/w_1(x, t). \quad (1)$$

In expression (1), the subscript refers to the partial derivative of $w(x, t)$ with respect to its first argument.

To see that the value of such a hedged position does not depend on the price of the stock, note that the ratio of the change in the option value to the change in the stock price, when the change in the stock price is small, is $w_1(x, t)$. To a first approximation, if the stock price changes by an amount Δx , the option price will change by an amount $w_1(x, t) \Delta x$, and the number of options given by expression (1) will change by an amount Δx . Thus, the change in the value of a long position in the stock will be approximately offset by the change in value of a short position in $1/w_1$ options.

As the variables x and t change, the number of options to be sold short to create a hedged position with one share of stock changes. If the hedge is maintained continuously, then the approximations mentioned above become exact, and the return on the hedged position is completely independent of the change in the value of the stock. In fact, the return on the hedged position becomes certain.³

Main Idea

- **Consider a portfolio with stocks and options**
- **When stock goes up, you lose money on the options and vice versa**
- **Such a portfolio (if perfectly balanced) has no risk**
- **Thus the portfolio increases at the risk-free interest rate, “ r ”**

Discuss with your neighbor the main idea.

Convince yourselves that it is both true and nontrivial.

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³ This was pointed out to us by Robert Merton.

Derivation

In general, since the hedged position contains one share of stock long and $1/w_1$ options short, the value of the equity in the position is:

$$x - w/w_1. \quad (2)$$

The change in the value of the equity in a short interval Δt is:

$$\Delta x - \Delta w/w_1. \quad (3)$$

Assuming that the short position is changed continuously, we can use stochastic calculus⁵ to expand Δw , which is $w(x + \Delta x, t + \Delta t) - w(x, t)$, as follows:

$$\Delta w = w_1 \Delta x + \frac{1}{2} w_{11} v^2 x^2 \Delta t + w_2 \Delta t. \quad (4)$$

Requires Stochastic Calculus

NOTE: This assumes rates of stock change obey a random walk

Black-Scholes Equation

Dropping the Δt from both sides, and rearranging, we have a differential equation for the value of the option.

$$w_2 = rw - rxw_1 - \frac{1}{2} v^2 x^2 w_{11}. \quad (7)$$

where $w_2 = \frac{\partial w}{\partial t}$, $w_1 = \frac{\partial w}{\partial x}$, $w_{11} = \frac{\partial^2 w}{\partial x^2}$

$$\frac{\partial w}{\partial t} + \frac{1}{2} v^2 x^2 \frac{\partial^2 w}{\partial x^2} = r \left(w - x \frac{\partial w}{\partial x} \right)$$

A sort-of-diffusion equation

Writing t^* for the maturity date of the option, and c for the exercise price, we know that:

$$\begin{aligned} w(x, t^*) &= x - c, & x &\geq c \\ &= 0, & x &< c. \end{aligned} \tag{8}$$

change variables and solve...

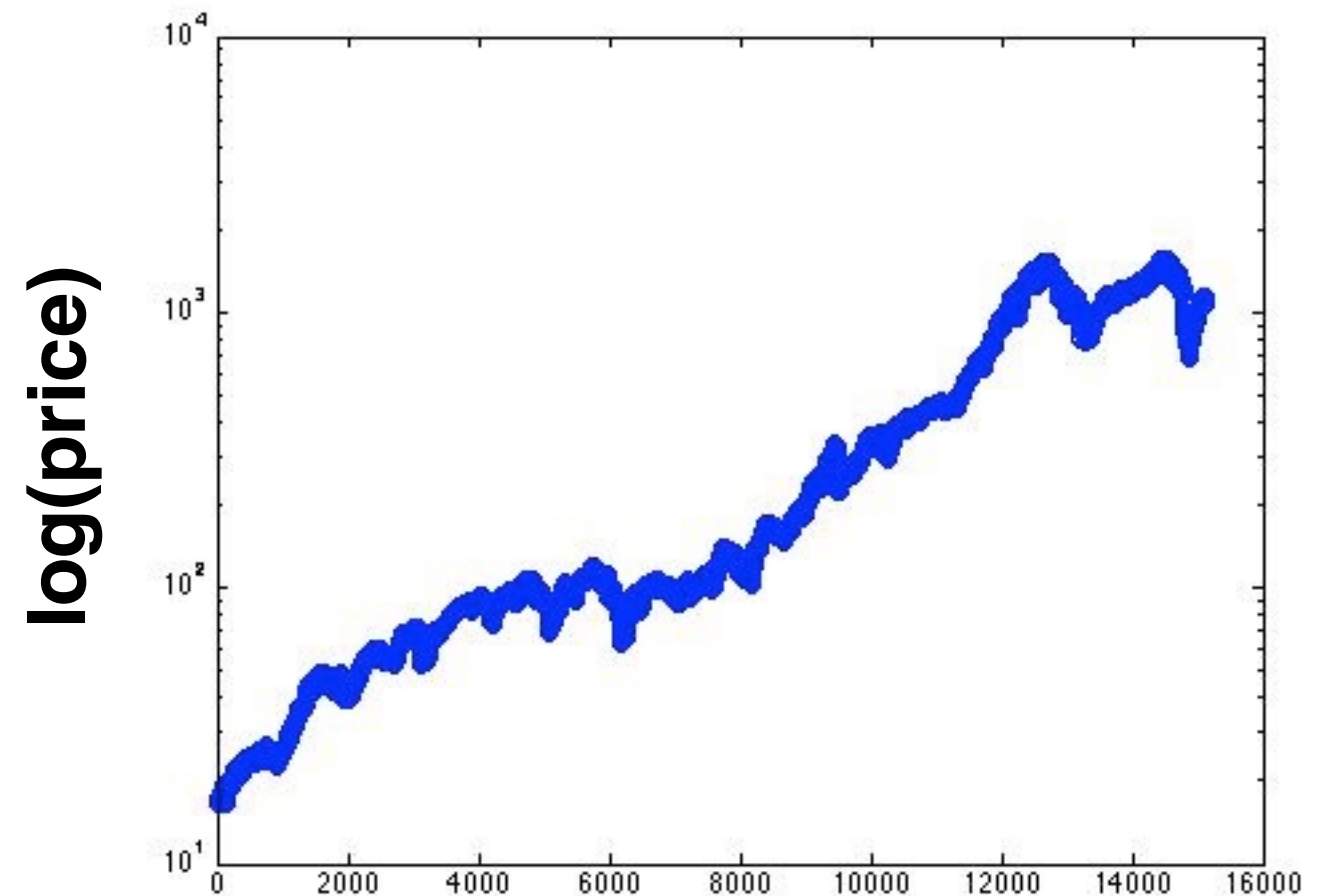
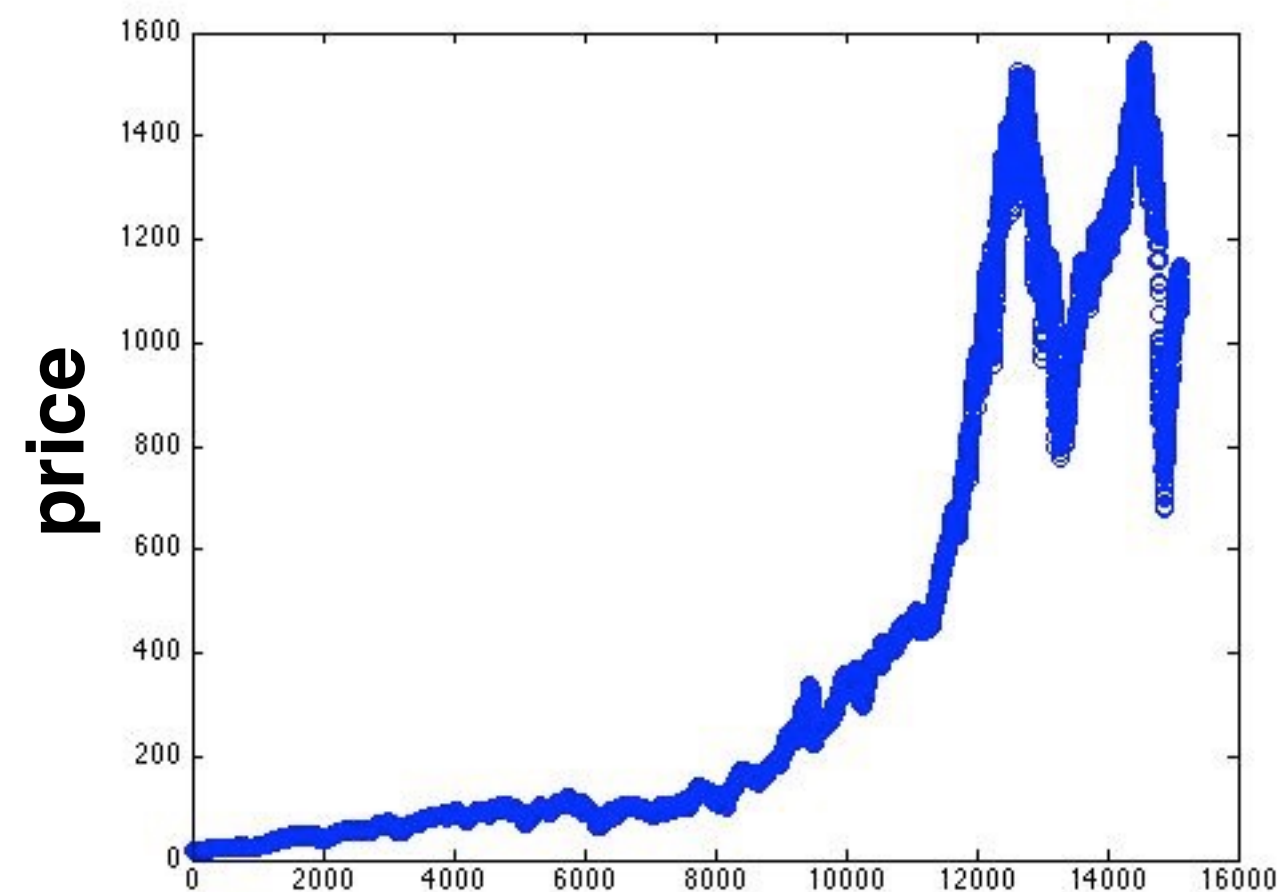
The differential equation (10) is the heat-transfer equation of physics, and its solution is given by Churchill (1963, p. 155). In our notation, the solution is:

$$y(u, s) = 1/\sqrt{2\pi} \int_{-u/\sqrt{2s}}^{\infty} c \left[e^{(u + q\sqrt{2s}) \left(\frac{1}{2} v^2 \right) / \left(r - \frac{1}{2} v^2 \right)} - 1 \right] e^{-q^2/2} dq. \tag{12}$$

Solution to equations like this covered in AM 105

S&P 500 Prices

**Central question in lab and homework:
do stock prices follow random walks?**



16000 days (1953–present)